

Heavy-State, Active-Closure, and Charged-Lepton-Flavour Extension of the Neutrino Standing-Carrier Space

Continuum survival under direct-state, weak-universality, and charged-lepton-flavour constraints

PCR-5D3

Principal result

Relative to the fixed successor target THUCL-D3-1.1, a source-first one-singlet family indexed by $M \in [28, 30]$ GeV realizes the direct-state, active-closure, and charged-lepton-flavour records on one physical point while preserving its oscillation standing core. Distinct masses give distinct threshold and oriented-matching records in the retained PCR-5B packet. Consequently,

$$|\mathrm{im} \rho_{D3}| \geq \mathfrak{c}, \quad \left| Q\left(T_{D3}^{HUL,1.1}\right) \right| \geq \mathfrak{c}.$$

A separate finite-submatrix family gives an exact marginal active-closure obstruction on a compact deficit interval. No complete-fibre elimination, proper-base-image theorem, or fixed-base splitting theorem is asserted.

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Abstract

PCR-5B identifies the standing-carrier quotient of the fixed neutrino-oscillation target, and PCR-5C supplies the functorial calculus of restriction and lift fibres under target extension. This paper extends that target by direct heavy-neutral-state searches, operational active-sector closure and weak-universality records, and the charged-lepton-flavour process $\mu \rightarrow e\gamma$. The active-closure component is the expressly versioned successor target TU-D3-1.1; the resulting joint target is THUCL-D3-1.1.

A source-first Weinberg-plus-one-singlet family is constructed for $M \in [28, 30]$ GeV, with no added charged-lepton-flavour operator. The family lies in the frozen ATLAS prompt-response specialization with positive mass, lifetime, and mixing margins; it passes the applicable atmospheric/reactor and low-mass weak-universality constraints; and its heavy contribution to $\mu \rightarrow e\gamma$ vanishes because $U_{\mu N} = 0$, leaving only the negligible light-neutrino GIM response. All constraints are evaluated on one source-coherent physical apex. The complete retained oscillation packet is recovered in situ, with the enriched cells still present.

Unequal masses change the physical heavy threshold and oriented matching datum already retained by the PCR-5B standing packet. The interval therefore injects into the restriction image, proving

$$|\mathrm{im} \rho_{D3}| \geq \mathfrak{c}, \quad \left| Q\left(T_{D3}^{HUL,1.1}\right) \right| \geq \mathfrak{c}.$$

A pure finite-submatrix family is also shown to have an empty accepted locus in one applicable active-closure chart for $\delta \in [0.02, 0.03]$, without implying complete base-fibre elimination. Candidate same-base CLFV refinements are dispositioned separately: the complete Warsaw dipole and a universal electromagnetic dipole alter the old source-facing trace, while a protocol-indexed localized environment proposal does not satisfy the constitutive source-coherence burden. Thus fixed-base splitting, properness of the base image, and the complete D3 atlas remain open.

Keywords. AASC; neutrino oscillation; heavy neutral lepton; nonunitarity; lepton-flavour universality; charged-lepton flavour violation; standing carrier; target extension; homotopy lift fibre; source-first realization.

Principal theorem

Let $\hat{T}_{D3}^{HUL} = \text{THUCL-D3-1.1}$ and let

$$R_{D3} : \mathcal{N}(\hat{T}_{D3}^{HUL}) \longrightarrow \mathcal{N}(T_{\text{osc}}), \quad \rho_{D3} = \pi_0(R_{D3})$$

be the retained standing-core restriction, with connected-class fibres

$$\mathcal{L}_{D3}(q) = \mathcal{N}(\hat{T}_{D3}^{HUL}) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].$$

On every transparent physical chart the frozen oscillation evaluator is recovered with all D3 cells present:

$$\Phi_4^{\text{ret},D3}(W_{H,\zeta}) = \Phi_4(W_H), \quad \Xi_{\text{osc}}^{\text{ret},D3}(W_{H,\zeta}) = \Xi_{\text{osc}}(W_H).$$

For each $M \in [28, 30]$ GeV, the source-first prompt-HNL family supplies a joint H/U/L survivor in the declared ATLAS response specialization and the fixed TU-D3-1.1 active-closure target. Distinct masses have distinct retained threshold and oriented-matching data. Hence

$$\boxed{|\mathrm{im} \rho_{D3}| \geq \mathfrak{c}}, \quad \boxed{\left| Q(\hat{T}_{D3}^{HUL}) \right| \geq \mathfrak{c}}.$$

On the complete pure finite-submatrix chart,

$$Z_{U,\text{sub}}^{\text{pure}}(\delta) = \emptyset, \quad \delta \in [0.02, 0.03],$$

which is a marginal active-closure obstruction, not a theorem of complete-fibre elimination. The full support of $\text{im } \rho_{D3}$, fixed-base splitting, properness, and the global D3 atlas are not determined here.

Scope of the theorem

The result is relative to the fixed targets TH-D3-1, TU-D3-1.1, TL-D3-1, and THUCL-D3-1.1; the PCR-3 finite source comparison class; PCR-4 one-point physiology; the PCR-5B/B15 base atlas; PCR-5C target-extension calculus; and the D2 in-situ transparency discipline. The low-mass universality component is a separately declared target, not a reconstructed collaboration likelihood. The prompt-lifetime equation is used as the frozen ATLAS signal-response specialization and is accompanied by a positive applicability margin. No unique HNL, unique see-saw, complete D3 classification, proper base image, or concrete same-base fibre multiplicity is claimed.

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1 Introduction and principal result

1.1 From the oscillation carrier space to the D3 target

The D3 target asks which already closed oscillation standing-carrier classes admit at least one lawful realization of the direct-state, active-closure, and charged-lepton-flavour records on one physical point. The object is not a preferred HNL model. It is the restriction map and its fibres:

$$R_{D3} : \mathcal{N}(\widehat{T}_{D3}^{\text{HUL}}) \rightarrow \mathcal{N}(T_{\text{osc}}), \quad \mathcal{L}_{D3}(q) = \mathcal{N}(\widehat{T}_{D3}^{\text{HUL}}) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].$$

A base class can be eliminated only if every framed D3 lift fails a named active target relation, is undefined, is response-out-of-scope at the declared comparison resolution, or requires a source/target update. One failed seesaw point, one absent particle name, or one null search cannot establish elimination.

1.2 Theorem strategy

The proof proceeds in the fixed order

$$\begin{aligned} &\text{target/source freeze} \Rightarrow \text{source-complete raw lifts} \Rightarrow \text{one-point physicality} \\ &\quad \Rightarrow \text{in-situ base transparency} \Rightarrow \text{SCIC restriction} \\ &\quad \Rightarrow \text{marginal/joint fibres} \Rightarrow \text{base image and coordinate diagnosis.} \end{aligned}$$

No empirical response is used in source formation. No source deletion defines target restriction. Standing-coordinate terminology is introduced only after fibre membership has been determined.

Throughout the paper, *explicit* denotes the source families and overlap loci constructed in full; it does not denote a provisional or preferred physical class.

1.3 Summary of conclusions

The analysis establishes three results on explicit, source-coherent loci. First, the prompt-HNL family indexed by $M \in [28, 30]$ GeV survives the simultaneous direct-state, active-closure, and CLFV target. Second, the physical threshold and oriented matching datum separate the retained PCR-5B classes along that interval, giving continuum cardinality in the restriction image and hence in the enriched quotient. Third, a distinct pure finite-submatrix family fails one applicable U chart on a compact deficit interval; this is a marginal obstruction and is not promoted to base selection.

Principal conclusions

Question	Conclusion	Scope
Does the joint D3 target have survivors?	Yes: the law-only class and a continuum prompt-HNL family.	Established
How large is the surviving base image?	$ \text{im } \rho_{D3} \geq \mathfrak{c}$.	Established lower bound
How large is the enriched quotient?	$ Q(\hat{T}_{D3}^{\text{HUL}}) \geq \mathfrak{c}$.	Established lower bound
Does the pure finite-submatrix chart survive U?	No accepted point for $\delta \in [0.02, 0.03]$.	Marginal chart result
Is the base image proper?	Not determined.	Open
Is there a concrete continuum inside one fixed fibre?	Not determined.	Open
Is the complete D3 atlas known?	No; the theorem is restricted to the explicit loci analysed below.	Open

2 Upstream structure and no-backflow discipline

2.1 The oscillation standing-carrier base

PCR-5B supplies an exact groupoid-valued atlas

$$\mathcal{A}_{\text{osc}}^{\text{exec}} \simeq \mathcal{N}_{5, \text{osc}}^{\text{exec}}, \quad \pi_0(\mathcal{A}_{\text{osc}}^{\text{exec}}) \cong Q_{\text{osc}}^{\text{SCIC}},$$

with LAW, LOCAL, HID, COMP, SOL, and HYB source charts, all pairwise and finite higher overlaps, framed provenance, typed failure frontiers, and the B15 target-extension datum. The base packet is

$$\Xi_{\text{osc}} = (T_{\text{face}}, R_{\text{stand}}, \Delta_{\text{fr}}^{\text{del}}, C_{\text{act}}, M_{\text{match}}, B_{\text{bdy}}, F_{\text{same}}).$$

Distinct base components remain distinct in every genuine extension because their separating old-target witnesses are retained.

2.2 Target-extension calculus

PCR-5C supplies base-source conservativity, connected-subgroupoid homotopy fibres, lift pseudofunctors, Grothendieck reconstruction, no merger, path equivalence over the base, parameterized fibres, one-point joint-target formation, and failure provenance. The present analysis specializes those theorems without modifying their hypotheses or conclusions.

2.3 D1 and D2 as controls

D1 establishes that absolute kinematic scale is not an oscillation-carrier selector. D2 establishes that the frozen ^{76}Ge response adds a conjugation/LNV coordinate while preserving the full base. D2 also proves the exact formation rule used below: the frozen

evaluator is applied with the new process present, and only afterward may a deletion/retraction theorem be stated. D1 and D2 are comparison controls, not selectors for D3.

2.4 MGN5, MGN6, and low-energy/UV separation

MGN5 supplies the rule that matrix size, eigenvalue labels, and threshold notation do not carry standing unless they factor through a witness-bearing quotient-reduced mass carrier. MGN6 supplies the common-interface and quotient rules for mixing, normalization, phase, and boundary-transition data. The low-energy exact-character result supplies the ultraviolet nonpromotion boundary: matching to the Weinberg class does not establish a heavy state, while heavy-state production, nonunitarity, and CLFV can distinguish ultraviolet continuations when they are independently witnessed.

Theorem 2.1 (No-backflow under the D3 target extension). *No D3 empirical limit, chart label, source family, direct-state interpretation, or standing-coordinate diagnosis may alter PCR-5B survivor membership, the seven-channel base packet, B15 provenance, or SCIC equivalence. An enriched incidence that changes an old-target output is routed by its in-situ packet to the actual base component or to source/target update.*

Proof. The old target, witnesses, continuations, and lawful equivalence are embedded in the D3 target. PCR-5C no-merger and SCIC witness completeness force every old-target change to be visible. A D3 label cannot suppress that witness without becoming a second classifier or target replacement. $\square$

3 D3 target and comparison class

3.1 Target specification

The operative frozen objects are

$$T_H = \text{TH-D3-1}, \quad T_U = \text{TU-D3-1.1}, \quad T_L = \text{TL-D3-1}, \quad \hat{T}_{D3}^{\text{HUL}} = \text{THUCL-D3-1.1}.$$

The target specification fixes publication/data identity, units, mass/lifetime/topology domains, response-status grammar, correlation policy, and update boundary. The separate TU-D3-1.1 addendum replaces only the U and joint identifiers by adjoining the low-mass five-record LFU response object before any D3 source is evaluated. Its dependency is

$$\text{primary LFU record} \longrightarrow \text{TU-D3-1.1 freeze} \longrightarrow \text{D3 source evaluation}.$$

The addendum is independent of the D3 endpoint: it contains no 28–30 GeV source, no HNL mixing, no survivor statement, and no cardinality calculation. New HNL searches, later MEG II data, other CLFV channels, positive signals, cosmology, symmetry records, or any change to the LFU vector, covariance, or 95-percent acceptance rule are target updates.

Theorem 3.1 (Target specificity of the low-mass universality extension). *Every theorem using the five-record low-mass LFU chart is relative to TU-D3-1.1 and THUCL-D3-1.1. The identifiers TU-D3-1 and THUCL-D3-1 do not contain that covariance or acceptance relation. No source evaluation is permitted to backflow into the addendum.*

3.2 Carrier-neutral response bands

A direct-state object is the complete production-propagation-decay instrument after response transport. The active-closure object is

$$\mathfrak{C}_{\text{act}} = (\text{Norm}, \text{Leak}, \text{ZeroDist}, \text{Universality}, \text{EW}, \text{CC/NC}, \text{Cov}, \text{Cont}),$$

with $N = (I - \eta)U$ and $\eta = \frac{1}{2}\Theta\Theta^\dagger$ only on their proved charts. The CLFV object is the source-defined $\mu \rightarrow e\gamma$ transition transported through the MEG II event instrument and acceptance relation.

3.3 Applicability and response status

Every chart relation has exactly one status

$$\{\text{N/A}, \text{Pass}, \text{Fail}, \text{Undef}, \text{ROOS}, \text{SU}, \text{TU}\}.$$

Applicability is fixed before truth evaluation. Null observation is not physical zero; out-of-scope response is not exclusion; missing response is not N/A; new source or target structure is not a point.

3.4 Source reduct

For a valid enriched source point X over a B15 frame H_a ,

$$U_{D3}(X) = \text{ActRed}_{T_{\text{osc}}}(\text{Forget}_{D3/T_{\text{osc}}} X) \simeq_{\text{src}} H_a.$$

The source-landing theorem shows that every D3 architecture admitted here lies in the PCR-3 source theory; no additional source extension is required. Detector response and likelihood are target/evaluator data, not source primitives.

4 Source-complete D3 candidate space

4.1 Source presentations and parameter domains

For each B15 skeleton $a = (F, \sigma)$, write

$$H_a = (\Theta_a^{\text{raw}}, E_a, A_a, \text{Stat}_a, Z_a, \Gamma_a, p_a, M_a, \text{Prov}_a, \Delta_{\text{fr}}^{\text{del}}, \partial\text{Fail}_a, \text{Sep}_a, \text{Ov}_a).$$

The raw D3 parameter object is

$$\Theta_{a,D3}^{\text{raw}} = \Theta_a^{\text{raw}} \times \Theta_{a,H} \times \Theta_{a,U} \times \Theta_{a,L} \times \Theta_{a,\text{cross}}.$$

The D3 factors are generated before response evaluation from existing local, hidden, law/process, composite, solitonic, boundary, environmental, and finite-hybrid source constructors.

4.2 Applicability and residual equations

Define

$$E_{a,D3} = E_a \cup E_{a,H} \cup E_{a,U} \cup E_{a,L} \cup E_{a,\text{match}} \cup E_{a,\text{bdy}} \cup E_{a,\text{cross}} \cup E_{a,\text{higher}} \cup E_{a,\text{cone}}.$$

Each equation is preceded by its source/target applicability predicate. Active equations return pass/fail/undefined; theoremmically absent equations return N/A; response-regime failure returns ROOS. Source and target updates are disjoint exits.

Definition 4.1 (Solved D3 chart). The solved chart is

$$Z_{a,D3} = \{(\theta, \zeta) \in \Theta_{a,D3}^{\text{raw}} : E_{a,D3}(\theta, \zeta) = 0, \text{Acc}_{HUL}(\theta, \zeta) = 1\}.$$

Its lawful presentation groupoid $\Gamma_{a,D3}$ is generated by the inherited B15 arrows and the applicable D3 source, response, threshold, matching, detector-chart, and framed-provenance arrows that preserve the complete enriched packet. Here $\text{Acc}_{HUL} = 1$ means that every active, applicable response predicate passes; theoremtically absent and response-out-of-scope charts retain their certified N/A or ROOS status and do not masquerade as either pass or exclusion.

Theorem 4.2 (Source coverage and pre-candidate disposition). *The D3 source calculus has two stages.*

1. *The union of solved source-coherent D3 charts over all B15 skeletons and lawful finite hybrids covers every source-coherent D3 realization in the frozen comparison class.*
2. *Every raw typed proposal receives exactly one total source disposition before candidate formation. Proposals satisfying the PCR-3 constitutive source equations enter the solved cover; source-incoherent proposals are assigned SourceMalformed, an explicit source update, or NotReady, and are excluded from the solved candidate cover.*

No fitted output section or post-hoc response correction is included.

Proof. PCR-3 source completeness and the executed source-landing matrix cover resolved local states, hidden dilations, finite production/decay processes, boundary effects, operational closure, CLFV transitions, and arbitrary finite hybrids after constitutive source coherence is imposed. Target-only data remain evaluator coordinates. Typed terms that fail their source equations do not become candidates by carrying recognized family labels. In particular, the protocol-indexed MEG-localized proposal lacks the required environment law and is assigned SourceMalformed / NotReady before the solved cover. J2-type arbitrary fitted response sections remain excluded by the source freeze. $\square$

5 D3 response equations and familywise specialization

5.1 Universal response system

For each B15 skeleton $a = (F, \sigma)$, the raw variables are the inherited base point θ , direct-state process data ζ_H , closure data ζ_U , CLFV law/process data ζ_L , and common-apex data ζ_X . Family names do not determine survival. For each source presentation, survival is the solution object of the following applicability-guarded relations.

ID	Object	Exact requirement
S00	Inherited source/base	$\theta \in Z_a$; all PCR-3 source equations, PCR-4 residuals, B15 applicability data, and the framed old core hold.
S01	D3 source reduct	$\text{ActRed}_{T_{\text{osc}}} \text{Forget}_{D3/T_{\text{osc}}}(\theta, \zeta_H, \zeta_U, \zeta_L, \zeta_X) \simeq_{\text{src}} H_a$; no retuning or replacement.
H00	Instrument applicability	For every frozen direct-search instrument j , mass, lifetime, flavour, topology, and release assumptions return exactly one of applicable, ROOS, N/A, or update.
H01	Physical process	Applicable production, propagation, decay, displacement, and final-state maps form one source-coherent completely positive event instrument.
H02	Common source parameters	Every H response uses the same mass spectrum, mixing/attachment, lifetime, threshold, and matching data carried by the source point.
H03	Detector transport	The source process is transported through the exact frozen efficiency/response object; no plotted-limit interpolation replaces the declared acceptance relation.
H04	H acceptance	Every active applicable H instrument passes its publication-defined likelihood or exclusion predicate.
H05	H complement	First failure is localized as process, topology, response, likelihood, undefined, ROOS, source update, or target update.
U00	U chart routing	Operational, finite-submatrix, heavy-EFT, boundary, and open-system charts are partitioned before numerical evaluation; no generic channel is forced into a matrix chart.
U01	Operational packet	$\mathfrak{C}_{\text{act}} = (\text{Norm}, \text{Leak}, \text{ZeroDist}, \text{Universality}, \text{EW}, \text{CC/NC}, \text{Cov}, \text{Cont})$ is evaluated on the complete physical apex.
U02	Specialized equation	On an applicable finite-submatrix chart, N is the actual active block and $NN^\dagger \preceq I$; on a heavy-EFT chart, matching and truncation conditions are proved before $\eta = \frac{1}{2}\Theta\Theta^\dagger$ is used.
U03	U acceptance/covariance	The applicable public-data or global-fit acceptance is evaluated with its frozen covariance/correlation policy; overlapping fits are not multiplied.
U04	U continuation	Every licensed matter, NC, boundary, threshold, and RG/EFT continuation preserves the declared chart and common source identity.
U05	U complement	First failure is localized as operational closure, normalization, zero-distance, universality/EW, covariance, undefined, ROOS, or update.
L00	CLFV source routing	Every CLFV source is an admitted PCR-3 process or finite hybrid with coefficient formed before response evaluation. The complete Warsaw chart and the target-scale electromagnetic chart are distinct source objects.
L01	Matching/output grades	On every admitted source-coherent electroweak or target-scale chart, matching retains every photon, Z, H, charged-current, auxiliary, and boundary descendant and evaluates its old/H/U/L grades. The protocol-indexed MEG-localized proposal is not an admitted chart: without a constitutive physical environment law it receives <code>SourceMalformed</code> / <code>open</code> before any solved output grade is formed.
L02	RG/continuation	Continuations are evaluated only on admitted source-coherent charts and must retain every generated old/H/U/L output. The MEG-localized proposal has no licensed support-preserving continuation algebra and is excluded from the admitted comparison class.

ID	Object	Exact requirement
L03	Physical amplitude	The complete target-scale transition amplitude is derived from the admitted source operator and its physical matrix element; phase-space normalization and event transport are formed afterward. An auxiliary colligation may represent only an already source-derived amplitude.
L04	Event response	The amplitude is transported through the frozen MEG II signal/background instrument and publication-defined likelihood.
L05	L acceptance	The active L response lies in the frozen 90% C.L. acceptance; exact zero is used only when source-defined.
X00	One source identity	H, U, and L are projections of one source-coherent point; separately fitted parameters are not spliced.
X01	Mass/rank	Direct mass, active spectrum, rank/nullity, thresholds, and effective matching agree on all common uses.
X02	Flavour attachment	Charged-current boundary, active-heavy attachment, and CLFV flavour indices are one common typed datum.
X03	Lifetime/continuation	H prompt/displaced lifetime and U decay/leakage continuation agree.
X04	Matching	Direct, EFT, electroweak-dipole, target-scale process, and loop charts share oriented matching only where jointly applicable; no UV origin is inferred for the explicit electromagnetic process.
X05	Boundary/statistics	Source/detector, near-detector, event-instrument, nuisance, and covariance records are compatible rather than freely factored.
X06	Gauge/anomaly/vacuum	One representation, charge, anomaly, vacuum, and deformation branch supports the full point.
X07	Higher incidence	Every active three-way or higher source relation is represented before quotienting and satisfies its constitutive equations.

X08 - common cone. The inherited and D3 diagrams admit one physical common-cone apex; a tuple that is only pairwise compatible fails X08.

T00 - in-situ base routing. With all D3 cells physically present, the full old evaluator returns exactly one of: the framed base packet, another exact PCR-5B packet, source update, or target update. This routing clause is evaluated after S/H/U/L/X formation and before restriction descends.

Let $\text{Res}_{a,D3}$ be the ordered tuple of these residuals after applicability routing. Define

$$Z_{a,D3} = \{x \in \Theta_{a,D3}^{\text{raw}} : \text{Res}_{a,D3}(x) = 0, \text{Acc}_{HUL}(x) = 1\}.$$

The first-failure map $\text{FirstFail}_{a,D3}$ records the earliest active nonzero residual in the displayed order; if no equation is applicable it retains the certified N/A or ROOS chart rather than laundering it into pass.

5.2 Familywise specializations

For each concrete B15 skeleton, the D3 response system assigns an exact raw domain, applicability vector, marginal and joint solution formula, lawful groupoid, and ordered failure frontier. The following table records the six familywise specializations. It does not assert that every member of the unbounded PCR-5B skeleton class has been solved individually.

Family	Source variables/equations	H/U/L response system	Lawful groupoid and isotropy	Exact complement / first separators
LAW	Finite law/process terms, operator order, memory/nonlocal kernel, boundary attachment, continuation and matching.	H production/decay process or certified zero/ROOS; operational U channel; Q_{eB} , other lawful L process, or exact zero; X00-X08.	Fixed-boundary gauge/basis, process-chart, scheme and lawful continuation arrows; isotropy stabilizes the complete law, response and failure packet.	Operator applicability, channel closure, memory, response order, boundary, L process, or first failed X residual.
LOCAL	Finite neutral-state spectrum, exact full mixing block, representations, Yukawa/mass tensors, production, decay, lifetime and threshold.	H instrument from the same mass/mixing/lifetime; applicable finite-submatrix or EFT U equations; HNL-loop/dipole/contact L equation; X00-X08.	Heavy-state unitary rotations within degenerate sectors, fixed active-boundary rephasings, chart and threshold maps preserving the full packet; isotropy is the stabilizer of spectrum and attachments.	Mass/lifetime/topology, rank, row/column normalization, EFT applicability, loop amplitude, threshold or matching.
HID	Minimal active hidden support, Stinespring/isometric dilation, projection, ownership, memory, hidden threshold and continuation.	H visible reduction/decay or zero; operational U loss/projection/closure; hidden-loop/effective L response or zero; X00-X08.	Source-authorized hidden isometries/Kraus maps, visible fixed-boundary arrows and chart maps; isotropy stabilizes minimal support and projection.	Hidden support, trace completion, projection, leakage, memory, ownership, response applicability or common cone.
COMP	Constituent diagram, derivation map, retained support, collective threshold, matching orientation and boundary incidence.	H constituent production/decay; derived U closure with constituents retained; constituent/collective L process; X00-X08.	Constituent isomorphisms and dual descriptions preserving derivation, retained support and complete packet; isotropy stabilizes the constituent diagram.	Constituent loss, derivation failure, threshold, response topology, matching, boundary or cross-incidence.
SOL	Persistent profile, topological/profile charge, deformation branch, endability, localized boundary and continuation.	H profile excitation/decay; profile-mediated U continuation; environment-coupled L transition; X00-X08.	Profile deformations and coordinate maps preserving topology, endability, boundary and responses; isotropy is the profile stabilizer.	Profile existence, endability, deformation, closure, H/L response, boundary or cone.
HYB	Arbitrary lawful finite decorated hypergraph built from all source sorts with Inflow/Fuse/CoCarry/Project/boundary and higher-arity relations.	All marginal equations evaluated on one finite hypergraph; no marginal source splice; all active higher relations and X00-X08.	Hypergraph isomorphisms, constituent/dilation/profile arrows and gluing automorphisms preserving the pointed enriched core; isotropy is the full core stabilizer.	First failed constituent, gluing, cross, higher, source-composition, matching, boundary, response or cone relation.

Theorem 5.1 (Totality of source-indexed D3 data). *For every concrete B15 skeleton datum $H_{F,\sigma}$, the constructor $\mathfrak{D}_{D3}$ returns a well-typed raw domain, applicability vector, marginal and joint residual systems, lawful groupoid presentation, and ordered status/failure frontier. Every source point is assigned a formal status in $\{N/A, \text{Pass}, \text{Fail}, \text{Undef}, \text{ROOS}, \text{SU}, \text{TU}\}$. This is a theorem about construction and typing; it does not assert that every resulting symbolic solution object has been explicitly classified or that the global codescent atlas is complete.*

Proof. PCR-3/B15 supplies the canonical skeleton code and inherited base datum. S00–S01 preserve that index. H00–H05, U00–U05, L00–L05, X00–X08, and T00 each have a frozen applicability predicate and status codomain, so the constructor is total as a typed record map. The six family rows are exhaustive routing rules, with HYB retaining every finite cross and higher incidence. Solving the resulting equations for the unbounded skeleton class is a separate burden and remains open outside the marked records. $\square$

Global-atlas boundary

The constructor

$$\begin{aligned} H_{F,\sigma} &\mapsto \mathfrak{D}_{D3}(H_{F,\sigma}) \\ &= (\Theta_{F,\sigma,D3}^{\text{raw}}, \text{App}_{F,\sigma,D3}, Z_{F,\sigma,H}, Z_{F,\sigma,U}, Z_{F,\sigma,L}, Z_{F,\sigma,HUL}, \\ &\quad \Gamma_{F,\sigma,D3}, \partial \text{Fail}_{F,\sigma,D3}). \end{aligned}$$

is exact and source-indexed. The corresponding supplementary data instantiate the source-coherent loci used in the theorems below. The unbounded PCR-5B skeleton class is not solved in full. Accordingly, the global D3 codescent atlas and exhaustive disposition of every $q \in Q_{\text{osc}}^{\text{SCIC}}$ remain open; the exact results concern the law-only and prompt-HNL base loci and the pure-submatrix marginal obstruction.

6 One-point physical survival

6.1 Common physical apex

A D3 survivor is one physical object $W \in \text{PhysMod}_{4,D3}^{\text{sem}}$ carrying the retained oscillation channel, the three D3 instruments, matching, threshold, boundary, vacuum/deformation, gauge/anomaly, and every active cross relation. Marginally acceptable objects cannot be spliced.

Theorem 6.1 (Constraint-preserving D3 realization). *Let $(\theta, \zeta) \in Z_{a,D3}$. Then the PCR-3 semantic interpretation and PCR-4 common-cone data construct a single physical apex $W_{a,\theta,\zeta}$ such that every inherited, marginal, matching, boundary, cross, higher, and common-cone residual vanishes. Conversely every semantic D3 survivor has a chart representative of this form.*

Proof. Source coherence fixes one active source diagram and one parameter object. PCR-4 survivor equations provide inherited physicality. The D3 residual system forces H/U/L response maps, shared mass/rank/flavour/lifetime data, oriented matching, and common boundary/vacuum data to factor through the same source point. Higher incidence is retained before quotienting. Soundness is direct; coverage follows from the source-complete chart theorem. $\square$

6.2 No marginal splicing

If H, U, and L fits require different masses, mixings, lifetimes, threshold maps, or boundary attachments, the cross/common-cone residual fails. Equal scalar limits from separate charts do not produce one D3 object.

6.3 Physical nulls

Exact zero H production, zero closure defect, or zero CLFV amplitude is lawful only when produced by the source and accepted by the target. Non-observation alone does not create the zero; undefined or out-of-scope response is not a zero.

7 In-situ retained-oscillation transparency

7.1 Relative grading and ideal

For a framed base realization W_H and an enriched point $W_{H,\zeta}$, grade source/process/evaluator cells by old-target or D3-only output sort. Let $I_{D3}(H, \zeta)$ be the closed ideal generated by D3-only state, process, response, matching, boundary, failure, and higher-incidence cells. A chart is transparent when no operation with a D3 input has an old-target output and every old continuation preserves the grading.

Lemma 7.1 (Annihilation of D3-only cells by retained evaluators). *For every primitive old-target evaluator e , every licensed old continuation u , and every transparent physical point,*

$$e_u^{\text{ret}, D3}(W_{H,\zeta}) = e_u(W_H), \quad e_u^{\text{ret}, D3}(I_{D3}(H, \zeta)) = 0.$$

Every mixed evaluator term with at least one D3-only input and an old-target output vanishes.

Proof. Induct on the frozen PCR-4/SCIC evaluator grammar. The primitive case is the no-old-output formation rule. Products and tensor contractions preserve the grading. Adjoints and functional calculus preserve the ideal. Finite higher incidence remains D3-graded because its output sort was fixed before response evaluation. Matching and failure continuations preserve the grading by target freeze. The no-repair deletion algebra is inherited literally, so a D3 deletion cannot retune an old source kind or recompute an old packet. $\square$

Theorem 7.2 (Retained-Oscillation Transparency and Base-Conservativity). *For every framed PCR-5B/B15 realization W_H and every transparent D3 physical point $W_{H,\zeta}$,*

$$\Phi_4^{\text{ret}, D3}(W_{H,\zeta}) = \Phi_4(W_H), \quad \Xi_{\text{osc}}^{\text{ret}, D3}(W_{H,\zeta}) = \Xi_{\text{osc}}(W_H).$$

Channel by channel,

$$\begin{aligned} T_{\text{face}}^{\text{ret}} &= T_{\text{face}}, & R_{\text{stand}}^{\text{ret}} &= R_{\text{stand}}, & (\Delta_{\text{fr}}^{\text{del}})^{\text{ret}} &= \Delta_{\text{fr}}^{\text{del}}, \\ C_{\text{act}}^{\text{ret}} &= C_{\text{act}}, & M_{\text{match}}^{\text{ret}} &= M_{\text{match}}, & B_{\text{bdy}}^{\text{ret}} &= B_{\text{bdy}}, & F_{\text{same}}^{\text{ret}} &= F_{\text{same}}. \end{aligned}$$

No equality is defined by deleting, retracting, retuning, or re-evaluating after removal of D3 structure.

Proof. Apply the annihilation lemma to the primitive factors generating the complete frozen evaluator and its seven packet recodings. Packet completeness then yields the stated equality. $\square$

7.2 Routing nontransparent points

A D3 cell that feeds an old output is evaluated in situ. If the old packet equals a valid PCR-5B core, the enriched point is framed over that actual core. If the old target or boundary changes, the status is target update. If a new base-active primitive appears, the status is source update. Nontransparent points are never silently discarded from source coverage and never called transparent corrections.

Corollary 7.3 (Restriction and no merger). *The in-situ retained packet defines a natural standing-core restriction $R_{D3} : \mathcal{N}(\hat{T}_{D3}^{\text{HUL}}) \rightarrow \mathcal{N}(T_{\text{osc}})$. D3-SCIC equivalent enriched points have the same retained base class. No enriched class restricts to two distinct base SCIC components.*

8 Exact marginal and joint lift geometry

8.1 Connected-class fibres

For $q \in Q_{\text{osc}}^{\text{SCIC}}$, define

$$\begin{aligned}\mathcal{L}_H(q) &= \mathcal{N}(T_H) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q], & \mathcal{L}_U(q) &= \mathcal{N}(T_U) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q], \\ \mathcal{L}_L(q) &= \mathcal{N}(T_L) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q], & \mathcal{L}_{D3}(q) &= \mathcal{N}(\hat{T}_{D3}^{\text{HUL}}) \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].\end{aligned}$$

The joint fibre is a common-point homotopy pullback, not a set intersection of marginal components.

8.2 Framed chart formula

For a representative $n \in \mathcal{N}(T_{\text{osc}})[q]$, let $Z_{a,X}(n)$ be the solved source/response object in band $X \in \{H, U, L, HUL\}$ with a frame $R_X(y) \cong n$. Then

$$\mathcal{L}_X(q) \simeq \int_{(a,n) \in \mathcal{I}(q)} [Z_{a,X}(n) / \Gamma_{a,X}(n)],$$

where $\mathcal{I}(q)$ is the full B15 framed provenance category over the connected base component. The integral retains base isotropy and source monodromy. Applying π_0 gives the exact coarse fibre.

Theorem 8.1 (Exact fibre decomposition). *Every lawful D3 survivor occurs in exactly one $\mathcal{L}_{D3}(q)$; every lawful raw lift is assigned a total status; and*

$$Q(\hat{T}_{D3}^{\text{HUL}}) \cong \coprod_{q \in Q_{\text{osc}}^{\text{SCIC}}} \pi_0 \mathcal{L}_{D3}(q).$$

The same statement holds marginally for H , U , and L .

Proof. Coverage is the candidate-completeness theorem. One-point survival defines the solved chart objects. Retained transparency and routing make restriction total on survivors. Connected base components are disjoint, so each enriched class belongs to one fibre. The lift pseudofunctor and Grothendieck reconstruction from PCR-5C give the groupoid equivalence; π_0 gives the coarse disjoint union. $\square$

9 Direct-state response

9.1 Production, propagation, and decay response

For each frozen instrument j , a source point induces

$$\text{Resp}_{H,j}(X) = \mathcal{D}_j \circ \mathcal{T}_j(\text{Prod}_X, \text{Prop}_X, \text{Decay}_X, \mu_H, \tau_H, \text{Final}_X),$$

followed by the collaboration-defined acceptance predicate $\text{Acc}_{H,j}$. Production-only and visible-decay charts retain different lifetime and topology semantics. Equal scalar limits do not identify complete H packets.

9.2 Null and decoupling loci

A source with no direct-state production has an exact zero H response and passes the null target when the zero is source-defined. On an applicable production chart, let $R_{H,j}(m, g)$ be the exact transported response. Source formation fixes $R_{H,j}(m, 0) = 0$, and the frozen response map is continuous in the source coupling on each compact chart domain. Since the publication-defined null-search acceptance contains a neighbourhood of zero response, continuity gives $\varepsilon_j(m) > 0$ such that $|g| < \varepsilon_j(m)$ is accepted. Compactness gives a uniform positive minimum on every closed subinterval on which the chart remains applicable. The corresponding mass, lifetime, topology, and response values are D3-standing exactly when distinguished by the target packet.

Proposition 9.1 (Nonemptiness of the direct-state marginal). *The law-only class q_W has an H lift with exact zero direct-state response. Every applicable source skeleton with a continuously decoupling H coupling has nonempty accepted H submoduli. H failure is therefore parameter- and source-specific; absence of a named HNL does not eliminate a base class.*

9.3 Direct/EFT separation

A direct state retains physical mass support, lifetime, production, decay, displacement, and event topology. An EFT chart retains operator order, coefficient, scale window, oriented matching, and truncation semantics. Matching is not identity, and a direct exclusion in one topology does not exclude every EFT realization of the same low-energy operator.

10 Active-sector closure and nonunitarity

10.1 Operational closure packet

The U evaluator is formed on the complete physical point from normalization, leakage, zero-distance, universality, EW, CC/NC, covariance, and continuation records. Finite matrix charts are specializations:

$$N = (I - \eta)U, \quad \eta = \frac{1}{2}\Theta\Theta^\dagger \quad \text{only on their proved heavy-neutrino/EFT loci.}$$

A generic open-system, dilation, projection, decay, or boundary source is evaluated through the operational packet and is never forced into a matrix chart.

10.2 Applicable response charts

Let $\mathcal{J}_U$ be the frozen U-chart index and

$$\text{Cov}_U(X) = \{j \in \mathcal{J}_U : \text{App}_{U,j}(X) = 1\}.$$

An enriched source has U Pass exactly when $\text{Cov}_U(X) \neq \emptyset$, every applicable chart passes, and every nonapplicable chart is assigned a certified N/A or ROOS status. ROOS is neither zero nor Pass.

The 2023 full electroweak/flavour likelihood is valid only on its published heavy/integrated-out response regime. The full 2023 electroweak/flavour likelihood is valid only in its published heavy or integrated-out response regime, so its high-scale numerical bounds are not applied to an on-shell 28–30 GeV state. The five-record low-mass LFU vector, covariance, response formula, and acceptance relation are instead fixed before source evaluation in the separate target amendment TU-D3-1.1, *Low-Mass Weak-Universality Target-Freeze Addendum*. The amendment contains no prompt-HNL mass, mixing, survivor, fibre, or cardinality conclusion. It fixes the primary-source identity, the five records below, the covariance convention, the source-independent applicability predicate, and the declared source-blind 95-percent Gaussian acceptance rule before any carrier is evaluated.

10.3 Applicable low-mass universality response

For $M > m_\tau$, the calibrated heavy state is absent from pion, kaon, muon, and tau decay phase space. The exact accessible-light-row norms are

$$(NN^\dagger)_{ee} = 1 - \epsilon, \quad (NN^\dagger)_{\mu\mu} = (NN^\dagger)_{\tau\tau} = 1.$$

Define

$$r_\epsilon = (1 - \epsilon)^{-1/2}$$

and the five-component low-energy universality prediction

$$\mathbf{r}(\epsilon) = (R_{\mu e}^\pi, R_{\mu e}^K, R_{\mu e}^\tau, R_{\tau\mu}^\pi, R_{\tau\mu}^K)^T = (r_\epsilon, r_\epsilon, r_\epsilon, 1, 1)^T.$$

The TU-D3-1.1 experimental vector is

$$\mathbf{r}_{\text{exp}} = (1.0010, 0.9978, 1.0018, 0.9964, 0.9857)^T$$

with covariance

$$\Sigma_{\text{LFU}} = 10^{-6} \begin{pmatrix} 0.81 & 0 & 0 & 0 & 0 \\ 0 & 3.24 & 0 & 0 & 0 \\ 0 & 0 & 1.96 & 0.1064 & -0.1092 \\ 0 & 0 & 0.1064 & 14.44 & 1.7784 \\ 0 & 0 & -0.1092 & 1.7784 & 60.84 \end{pmatrix}.$$

The pion and kaon electron/muon ratios come from independent averages. To avoid the source-identified full correlation among the three purely leptonic tau ratios, the target retains only $R_{\mu e}^\tau$ from that trio. The nonzero off-diagonal entries are the published correlations of that ratio with $R_{\tau\mu}^\pi$ and $R_{\tau\mu}^K$, together with the published hadronic-tau correlation. Define

$$\chi_{\text{LFU}}^2(\epsilon) = (\mathbf{r}(\epsilon) - \mathbf{r}_{\text{exp}})^T \Sigma_{\text{LFU}}^{-1} (\mathbf{r}(\epsilon) - \mathbf{r}_{\text{exp}})$$

and the frozen low-mass LFU acceptance by $\chi_{\text{LFU}}^2 \leq \chi_{5,0.95}^2 = 11.0704976935$. This is the source-blind target rule declared in the addendum, not a reproduced collaboration likelihood.

Theorem 10.1 (Low-mass LFU response and uniform U margin). *For $0 \leq \epsilon \leq 5.02 \times 10^{-6}$,*

$$8.437 < \chi_{\text{LFU}}^2(\epsilon) \leq 8.445 < 11.0704976935.$$

In the whitened response norm,

$$\inf_{\epsilon \in [0, 5.02 \times 10^{-6}]} d(\mathbf{e}_U^{\text{lowM}}(\epsilon), \partial \text{Acc}_U) \geq \sqrt{11.0704976935} - \sqrt{8.445} > 0.421.$$

Thus the complete low-mass LFU block has a uniform positive acceptance margin.

Proposition 10.2 (Completeness of the low-mass U response). *Relative to the frozen TU-D3-1.1 packet, the applicable quantitative response on the marked 28–30 GeV electron-row source consists of the public atmospheric/reactor closure chart and the five-observable LFU covariance block above. The ORCA tau-row chart, the combined high-scale/Z-pole EW likelihood, and the integrated-out heavy-EFT charts fail their declared source/scale applicability guards and are ROOS. A new direct low-mass Z-width or additional collider-EW likelihood would be TU, not an unrecorded completion of the present response.*

Proof. Only the first three entries depend on r_ϵ . The displayed quadratic form is a convex quadratic polynomial in r_ϵ , so its maximum on the compact interval occurs at an endpoint. Direct evaluation gives 8.444531 at $\epsilon = 0$ and 8.437146 at $\epsilon = 5.02 \times 10^{-6}$. The covariance matrix is positive definite, so the whitened-distance statement follows. $\square$

For the prompt-HNL source below, the complete disposition is:

Frozen U chart	Status	Exact reason
Source-independent operational closure	Pass	One physical apex supplies exact normalization, leakage, zero-distance, CC/NC, covariance, and continuation data; the quantitative low-mass LFU block is evaluated below.
Public atmospheric/reactor submatrix chart	Pass	$NN^\dagger = \text{diag}(1 - \epsilon, 1, 1)$, $N_e > 0.9999949 > 0.987$, the off-diagonal row closures vanish, and the remaining rows are unit normalized.
Low-mass pion/kaon/tau LFU chart	Pass	The five-component response and covariance frozen by TU-D3-1.1 give $\chi_{\text{LFU}}^2 \leq 8.445 < 11.0704976935$ uniformly.
ORCA6 α_{33} one-parameter likelihood	ROOS	The publication varies the tau-row coordinate on a different source path; the explicit source carries an electron-row deviation.
2023 global EW/flavour specialization	ROOS	The published combined likelihood assumes a heavy/integrated-out response regime and is not applied to an on-shell 28–30 GeV state. No numerical row from that likelihood is reused independently.
Dimension-five/dimension-six heavy-EFT charts	ROOS	$M < m_W$ and there is no integration-out scale separation. Direct-state, oscillation-closure, and low-mass LFU responses are retained instead.

Frozen U chart	Status	Exact reason
Any absent source-specific chart	N/A	The source lacks the corresponding topology, detector primitive, or operator branch by theorem rather than by failed evaluation.

10.4 Source-first prompt-HNL base continuum with $z = 0$

Fix the physical light spectrum $\hat{m}$, PMNS matrix U_ν , and one gauge-singlet Majorana carrier. Let

$$I_M = [28, 30] \text{ GeV}.$$

The raw source domain contains all $(M, \epsilon) \in I_M \times (0, 1)$ before target evaluation. For the frozen ATLAS signal-response specialization, write

$$\Gamma_A^{\text{resp}}(M, \epsilon) = \epsilon \hat{\Gamma}_A(M), \quad \hat{\Gamma}_A(M) > 0.$$

This is the exact lifetime map of the declared response chart; it is not asserted as an unrestricted exact full-theory total width of every realization of the displayed 4×4 source. After source formation, select the response-specialized prompt-lifetime stratum

$$\epsilon_p(M) = \frac{\hbar c}{(0.2 \text{ mm}) \hat{\Gamma}_A(M)}.$$

The explicit source map therefore lands exactly on the interior ATLAS lifetime-response point $c\tau_A = 0.2 \text{ mm}$; no downstream exclusion value is used to generate the source.

Let $e = (1, 0, 0)^T$, $s_M = \sqrt{\epsilon_p(M)}$, $c_M = \sqrt{1 - \epsilon_p(M)}$, and $D_M = \text{diag}(c_M, 1, 1)$. Define the exact unitary

$$\mathbb{U}_4(M) = \begin{pmatrix} D_M U_\nu & s_M e \\ -s_M e^T U_\nu & c_M \end{pmatrix}.$$

With $m_\nu = U_\nu^* \hat{m} U_\nu^\dagger$, the exact mass blocks are

$$\begin{aligned} m_L(M) &= D_M m_\nu D_M + \epsilon_p(M) M e e^T, \\ m_D(M) &= s_M (c_M M e - D_M m_\nu e), \\ m_R(M) &= \epsilon_p(M) e^T m_\nu e + c_M^2 M. \end{aligned}$$

The physical source is the gauge-invariant Weinberg-plus-one-singlet hybrid, and no additional CLFV operator is present:

$$z = 0.$$

A positive source decay channel gives the lower bound

$$\hat{\Gamma}_A(M) \geq \frac{G_F^2 M^5}{192 \pi^3} F\left(\frac{m_\mu^2}{M^2}\right),$$

where

$$F(x) = 1 - 8x + 8x^3 - x^4 - 12x^2 \log x.$$

On I_M , $F > 0.99988$, so

$$0 < \epsilon_p(M) \leq 2.51 \times 10^{-6}.$$

The ATLAS prompt result covers the frozen mass domain $[8, 65] \text{ GeV}$, contains the lifetime-reweighted response interval $c\tau_A \in [0.1, 1.0] \text{ mm}$, and excludes electron mixings

above 1.1×10^{-5} throughout 15–30 GeV. The explicit source has the strict acceptance margin

$$g_H = 1.1 \times 10^{-5} - \sup_{M \in I_M} \epsilon_p(M) > 0.$$

In the normalized continuous chart coordinates define

$$g_{H,\text{app}} = \inf_{M \in I_M} \min \left\{ \frac{M - 8 \text{ GeV}}{57 \text{ GeV}}, \frac{65 \text{ GeV} - M}{57 \text{ GeV}}, \frac{1.1 \times 10^{-5} - \epsilon_p(M)}{1.1 \times 10^{-5}} \right\} > 0.35.$$

The lifetime coordinate has exact membership in the declared ATLAS response specialization and lies strictly inside its reweighted lifetime interval. Define

$$g_{\tau,\text{app}} = \min\{0.2 - 0.1, 1.0 - 0.2\} \text{ mm} = 0.1 \text{ mm} > 0.$$

Thus the explicit source has a positive lifetime-domain applicability margin in addition to the positive mass/mixing margin. No unrestricted full-source lifetime identity is claimed.

The active block is

$$N_M = D_M U_\nu, \quad N_M N_M^\dagger = \text{diag}(1 - \epsilon_p(M), 1, 1).$$

The conditionally normalized visible oscillation kernel is exactly the PMNS kernel. The U chart table above gives Pass on every applicable chart. The public oscillation margin is

$$g_{U,\text{osc}} = \inf_{M \in I_M} (1 - \epsilon_p(M) - 0.987) > 0,$$

and the low-mass LFU response has the uniform Mahalanobis margin

$$g_{U,\text{LFU}} = \sqrt{11.0704976935} - \sup_{M \in I_M} \sqrt{\chi_{\text{LFU}}^2(\epsilon_p(M))} > 0.329.$$

Because $U_{\mu N} = 0$, every heavy contribution to $\mu \rightarrow e\gamma$ vanishes exactly. The remaining light-neutrino contribution satisfies the GIM bound

$$\text{BR}_\nu^{\text{light}}(\mu \rightarrow e\gamma) \leq \frac{3\alpha}{32\pi} \left(\frac{\Delta m_{\text{max}}^2}{m_W^2} \right)^2 < 6 \times 10^{-48}$$

using the deliberately conservative $\Delta m_{\text{max}}^2 < 1 \text{ eV}^2$. Therefore

$$g_L = 1.5 \times 10^{-13} - 6 \times 10^{-48} > 0.$$

Gauge singlets add no anomaly, the scalar vacuum is unchanged, the exact mass Lagrangian supplies one common physical apex, and the threshold/matching interval is compact.

Theorem 10.3 (Prompt-HNL one-point survivor continuum). *Let*

$$\mathcal{R}_p = \{(M, \epsilon_p(M), z = 0) : M \in I_M\}.$$

Every point of $\mathcal{R}_p$ is a source-coherent one-point H/U/L D3 survivor. Every H/U/L margin is a genuine Pass margin; no ROOS coordinate is used positively.

Proof. The exact block formulas give a lawful source and exact diagonalization. The source-to-response map lies exactly on the lifetime anchor of the frozen ATLAS response specialization and has the displayed positive mass/mixing applicability margin. The width bound places the complete mixing interval strictly below the ATLAS exclusion envelope. The public oscillation chart and the covariance-complete low-mass LFU chart pass with uniform margins, while every remaining frozen U chart is routed to its exact ROOS or N/A status. The heavy CLFV factor vanishes by flavour orthogonality and the light GIM response lies strictly below MEG II. The same Lagrangian, vacuum, boundary, and matching data define one common apex. $\square$

Theorem 10.4 (Continuum of surviving base classes). *Let $q_p(M) \in Q_{\text{osc}}^{\text{SCIC}}$ be the retained PCR-5B/B15 class of the source at mass M . Then*

$$\{q_p(M) : M \in I_M\} \subseteq \text{im } \rho_{D3}$$

and the classes are pairwise base-SCIC-distinct. Consequently

$$|\text{im } \rho_{D3}| \geq c.$$

Proof. The preceding theorem supplies one joint D3 lift over every M . Unequal masses change the physical singlet threshold and oriented matching record retained by the frozen M^{match} channel. These changes are neither basis, gauge, phase, nor chart arrows. The nondegenerate compact real interval therefore injects into the base image. $\square$

10.5 Complete pure finite-submatrix chart and exact marginal obstruction

Fix $M_{\text{hid}} = 1 \text{ TeV } I_3$, a source-determined lifetime, minimal charged-current source/detector attachment, matching at 100 GeV, and the inherited electroweak vacuum. Let $\delta \in [0.02, 0.03]$ and

$$\mathbb{U}_\delta = \begin{pmatrix} \sqrt{1-\delta} U_\nu & \sqrt{\delta} I_3 \\ -\sqrt{\delta} U_\nu & \sqrt{1-\delta} I_3 \end{pmatrix}.$$

Its active block is $N_\delta = \sqrt{1-\delta} U_\nu$, and the event-complete instrument

$$\mathcal{E}_{\lambda,\beta}^{(\delta)} = (1-\delta)\mathcal{E}_{\lambda,\beta}^0,$$

$$\mathcal{E}_{\lambda,\emptyset}^{(\delta)}(\rho) = \mathcal{E}_{\lambda,\emptyset}^0(\rho) + \delta \sum_{\beta} \text{Tr } \mathcal{E}_{\lambda,\beta}^0(\rho) |\emptyset\rangle\langle\emptyset|$$

is exactly trace preserving. The conditionally normalized visible kernel equals the old PMNS kernel.

Theorem 10.5 (Pure finite-submatrix obstruction). *On the complete pure finite-submatrix source, the public submatrix chart is applicable and*

$$Z_{U,\text{sub}}^{\text{pure}}(\delta) = \emptyset \quad \forall \delta \in [0.02, 0.03],$$

because $N_e = 1 - \delta \leq 0.98 < 0.987$. Its H leg is ROOS at 1 TeV, so this theorem is a marginal U obstruction only. A chart-changing boundary or HYB lift is evaluated in its actual U chart; no complete base-fibre elimination follows.

11 Charged-lepton-flavour response and source boundaries

11.1 Frozen event-response object

The L primitive is the MEG-II event-response packet for $\mu^+ \rightarrow e^+ \gamma$, including the published signal, background, nuisance, detector, likelihood, and acceptance semantics. A source point has L Pass when its complete source-defined amplitude and detector transport lie in the frozen accepted response set. Physical zero, a null result, and an unsupported mechanism are distinct statuses.

11.2 Old-trace audit of the Warsaw $Q_{eB}^{e\mu}$ source

The source-first Warsaw cell

$$Q_{eB}^{e\mu} = (\bar{\ell}_e \sigma^{\mu\nu} \mu_R) H B_{\mu\nu}$$

matches after electroweak breaking to photon and Z dipoles:

$$Q_{eB}^{e\mu} \mapsto \frac{v}{\sqrt{2}} \bar{e}_L \sigma^{\mu\nu} \mu_R (c_W F_{\mu\nu} - s_W Z_{\mu\nu}) + \dots$$

The Z-dipole term, together with the ordinary neutrino neutral current, produces the source process

$$\mu \longrightarrow e + Z^* \longrightarrow e + \nu_i + \bar{\nu}_i.$$

This is a neutrino-bearing source contribution and therefore enters the old target-facing source trace.

Theorem 11.1 (Old-trace disposition of the Warsaw dipole). *For the complete $Q_{eB}^{e\mu}$ source,*

$$E_{\text{old}}(Q_{eB}^{e\mu}) \neq 0.$$

The source is not an object of the same-base transparent CLFV subcategory. It is routed to the actual source-modified/HYB PCR-5B core determined by its in-situ old packet, or to target update if that packet is outside the frozen old comparison class.

Proof. The old source/boundary evaluator detects the nonzero $Z^* \rightarrow \nu\bar{\nu}$ incidence before any deletion. Smallness does not imply SCIC equality. A dimension-six Q_{eB}/Q_{eW} photon-aligned combination does not restore the theorem: the Q_{eW} component also carries charged- W dipole incidences. Any ultraviolet completion that cancels all such source outputs must be evaluated as a new intrinsic/HYB chart rather than assumed inside the present fibre. $\square$

Theorem 11.2 (Old-trace disposition of a universal electromagnetic dipole). *Let*

$$\mathcal{L}_\gamma^{\text{univ}}(c) = c \bar{e} \sigma^{\mu\nu} P_R \mu F_{\mu\nu} + \text{h.c.}$$

be physically active on any frozen old muon-parent source support. Then the source interaction changes the physical parent decay law and therefore has a nonzero old source-use trace on that support:

$$E_{\text{old}}^{\text{raw}}(\mathcal{L}_\gamma^{\text{univ}}(c)) \neq 0 \quad (c \neq 0).$$

It is not a calibration, nuisance, or lawful same-target redescription merely because the induced rate lies inside an uncertainty interval. The point is routed to its actual PCR-5B standing core or to target update. Only an independently source-environment-coupled process with support disjoint from every old protocol can enter the same-base theorem below.

Proof. The universal interaction adds a physical partial width wherever its source support contains a muon. PCR-1/PCR-2 lawful equivalence does not identify a changed source interaction with calibration transport. The in-situ old evaluator therefore detects the altered source law before any quotient. Numerical smallness affects enriched acceptance, not source identity. $\square$

11.3 Boundary of the same-base CLFV route

The attempted same-base route proposed a target-scale electromagnetic process multiplied by a profile that was nonzero on a protocol labelled MEG and zero on old, H, and U protocol labels. Protocol typing alone does not construct a PCR-3 environment carrier. The frozen sources do not provide a constitutive environment action or externally sourced boundary equation, a physical apparatus-response law acting uniformly across protocols, a complete vacuum/deformation solution, an environment-inclusive MEG event kernel, or an inherited-continuation audit.

Theorem 11.3 (Source-coherence boundary for a MEG-localized proposal). *Let $P_{\gamma, \text{MEG}}^{\text{raw}}$ denote the raw typed protocol-indexed proposal. At the present source scope,*

$$\text{Status}_{\text{src}}(P_{\gamma, \text{MEG}}^{\text{raw}}) = \text{SourceMalformed} \text{ or } \text{NotReady},$$

according to whether the missing constitutive environment law is treated as a failed source equation or as an explicit future source-extension burden. In either disposition the proposal is excluded before the solved D3 candidate cover and contributes no H/U/L solution object, no same-base fibre point, and no marked codescent node.

Proof. PCR-3 admits environmental and boundary carriers only after their constitutive equations, active support, lawful operations, and physical interpretation are fixed. The protocol-indexed profile supplies only a typed placeholder. It does not discharge source coherence, physical common-cone realization, detector response, or continuation closure. Therefore it cannot enter $Z_{a, D3}$ or any active source-indexed record. The rejected construction is retained only under `rejected_routes/` for provenance. $\square$

Same-base fibre boundary

No concrete exactly old-transparent CLFV source is currently proved. The manuscript therefore makes no claim that $\pi_0 \mathcal{L}_{D3}(q)$ is infinite or continuum-sized for any fixed base class. A future reopening must begin with a source-coherent physical environment law or another exact same-base source whose complete old/H/U/L/common-cone packet is calculated before response evaluation.

12 Simultaneous H/U/L realization

12.1 Joint solution object

For every B15 frame,

$$Z_{a, HUL} = Z_{a, H} \times_{\mathcal{C}_a}^h Z_{a, U} \times_{\mathcal{C}_a}^h Z_{a, L} \cap \{R_{\text{cross}} = R_{\text{higher}} = R_{\text{cone}} = 0\},$$

where $\mathcal{C}_a$ carries the shared mass/rank, flavour attachment, lifetime/continuation, matching, boundary, vacuum, gauge/anomaly, and statistics/correlation data. This is not a naive intersection of marginal accepted sets.

12.2 An explicit continuum of surviving base classes

Define

$$Q_{D3}^{\text{sem}} = \{q \in Q_{\text{osc}}^{\text{SCIC}} : \mathcal{L}_{D3}(q) \neq \emptyset\}.$$

Then $\text{im } \rho_{D3} = Q_{D3}^{\text{sem}}$ by the PCR-5C restriction theorem. The explicit calculation determines the subset

$$Q_{D3}^{\text{mark}} = \{q_W\} \cup \{q_p(M) : M \in I_M\} \subseteq Q_{D3}^{\text{sem}},$$

with $|Q_{D3}^{\text{mark}}| = \mathfrak{c}$. The pure finite-submatrix chart supplies a marginal U obstruction but no complete-fibre exclusion. No concrete same-base splitting support is established.

Theorem 12.1 (Continuum lower bound for the base image).

$$|\text{im } \rho_{D3}| \geq \mathfrak{c}, \quad |Q(\hat{T}_{D3}^{\text{HUL}})| \geq \mathfrak{c}.$$

No properness or exact-support theorem is claimed.

Proof. The prompt-HNL source family $\mathcal{R}_p \subset Z_{HUL}$ restricts to pairwise base-SCIC-distinct classes indexed by the nondegenerate interval I_M . The enriched quotient maps onto its base image. $\square$

12.3 No marginal inference

It remains possible for $\mathcal{L}_H(q), \mathcal{L}_U(q), \mathcal{L}_L(q)$ all to be nonempty while $\mathcal{L}_{D3}(q)$ is empty because the marginal lifts use incompatible sources, lifetimes, matching branches, or covariance assumptions. Conversely a mixed HYB point can survive jointly even when no pure family supplies the common source architecture.

13 Parameter descent and standing-visible coordinates

13.1 Parameter roles

Every parameter belongs, on each applicability stratum, to exactly one primary role:

1. source-defining and activity-bearing;
2. target-response or experimental nuisance;
3. lawful presentation, basis, scheme, or chart coordinate;
4. standing-visible enriched invariant or separator.

The same symbol can change role only across an explicit applicability or target-update boundary.

13.2 Mass, rank, and threshold

MGN5 requires rank, nullity, eigenstructure, and thresholds to be read on the quotient-reduced physical carrier. Raw matrix size, diagonalization labels, and fitted masses do not create standing. Direct-state mass support and lifetime are retained when the target detects them. Matching orientation and EFT scale windows remain distinct from direct-state identity.

13.3 Mixing and nonunitarity

MGN6 requires independently standing carriers or boundary projectors and a lawful common interface. A displayed N , η , or Θ has standing only through an admitted closure, normalization, zero-distance, universality, or transition witness. Rephasing, basis, permutation, degeneracy, and redundant chart directions descend.

13.4 CLFV nuisance and operator charts

Calibration, background, detector nuisance, and likelihood coordinates descend unless they change a standing target consequence. Dipole, HNL-loop, contact, and other sectors remain distinct when their process, continuation, applicability, or failure packets differ, even if one scalar upper limit agrees.

Theorem 13.1 (Parameter-descent closure). *The enriched packet descends through every lawful source/response groupoid, retains every target-visible threshold, closure, topology, CLFV, and failure difference, and admits no second classifier based on family name, naturalness, operator dimension, or preferred ultraviolet interpretation.*

14 No-repair source composition

14.1 Deletion profile

For a D3 survivor X with active source-block set $K(X)$, define

$$N_{D3,X} : 2^{K(X)} \rightarrow \{0, 1, \perp\}$$

by deleting the complement of a constitutively well-typed subset while holding all retained parameters fixed. No retuning, re-gluing, branch change, matching rewrite, replacement source, or post-deletion response fit is permitted.

14.2 S/R/P/Y taxonomy

Minimal sufficient subsets are coded as single-route S , redundant R , partially synergistic P , or fully synergistic Y . The code is derived from the deletion hypergraph, not from the source-family label.

14.3 Explicit source dispositions

Stratum	Typical code	Exact meaning
Law-only unitary null lift	S	The retained oscillation source alone realizes the null H/U/L target; deleting it destroys the base target.
Prompt in-window one-singlet mass continuum	S or P	One HYB source fixes the light spectrum and normalized kernel while the prompt heavy threshold varies; every H/U/L response is evaluated on the same source point.
Environment- localized CLFV proposal	SourceMalformed / open	The protocol-indexed raw term lacks a constitutive environment law and is excluded before the solved candidate cover.
HNL-induced H/U/L common chart	Y or P	Shared mass, mixing, lifetime, and loop data may be jointly necessary; deletion is evaluated without retuning cancellations.

Stratum	Typical code	Exact meaning
Hidden projection plus boundary normalization	P or Y	Projection and boundary structures can be jointly required for the U response even when either marginal source family survives elsewhere.

Theorem 14.1 (Source-composition exactness on explicit loci). *Every explicitly solved marked D3 stratum carries an exact no-repair sufficiency hypergraph and S/R/P/Y code. The uniform compiler defines the corresponding object for every B15 skeleton, but the complete source-organization atlas is not executed here.*

15 Standing-coordinate interpretation

15.1 Relative witness algebra

Remove the frozen base witnesses from the enriched witness algebra and denote the quotient-factorizing remainder by W_{D3}^{rel} . For enriched classes over the same base,

$$\text{Sep}_{D3}^{\text{rel}}(y_1, y_2) = \{(w, u) : w(u(y_1)) \neq w(u(y_2)), w \in W_{D3}^{\text{rel}}\}.$$

Minimality is claimed only where the primitive grammar is finite or well ordered; otherwise the exact nonempty separating family is exported.

15.2 Coordinate classes

D3 separators fall into active-sector closure, hidden dilation/projection, rank/threshold, source/detector continuation, direct-state process, CLFV transition, ultraviolet-only, or joint/higher-arity classes.

Theorem 15.1 (Relation to oscillation standing). *The proved marked loci exhibit:*

1. **Prompt direct-state base refinement:** *a continuum of genuine in-window HNL sources passes H/U/L and is separated at base resolution by threshold and oriented matching.*
2. **Pure-chart closure obstruction:** *the complete finite-submatrix source fails the applicable U row-normalization chart on $\delta \in [0.02, 0.03]$, without implying complete-fibre elimination.*
3. **Nontransparent electroweak dipole:** *the complete Warsaw $Q_{eB}^{e\mu}$ source has a nonzero old neutrino-source trace and is routed away from the same-base fibre theorem.*
4. **Environment-localized route rejected before candidate formation:** *protocol-indexed support does not define a source-coherent physical environment; the raw proposal is assigned SourceMalformed / NotReady and supplies no same-base separator.*
5. **Jointly unresolved general charts:** *arbitrary HNL-loop, boundary-modified, electroweak-dipole, and general HYB sources can tie H, U, and L through common mass, lifetime, threshold, and matching data. Their global source-indexed atlas remains open.*

Scope and interpretation

D3 establishes continuum base multiplicity and hence continuum enriched quotient multiplicity. It does not establish a proper base image, a concrete same-base CLFV direction, a universal CLFV attachment theorem, or a common generator. The Warsaw and universal dipoles are explicitly nontransparent. The protocol-indexed environment proposal is source-incomplete and excluded before candidate formation.

16 Overlap geometry on the explicit survivor locus

16.1 Pairwise and higher overlaps

For concrete B15 skeleton records $a = (F, \sigma)$ and $b = (G, \tau)$, the supplementary material constructor forms

$$\mathcal{O}_{ab} = \Gamma_{a,D3} \times_{\mathcal{N}_{D3}^{\text{core}}}^h \Gamma_{b,D3}$$

with the complete retained, H, U, L, boundary, matching, failure, source-composition, and common-cone equalizer. For a finite tuple J , it forms the direct higher object with pointed-core frames, cocycle equations, the complete response vector, and explicit $R_J^{\text{cone}} = R_J^{\text{higher}} = 0$.

16.2 Overlaps among the explicit survivor classes

Let $\mathcal{U}_{D3}^{\text{mark}}$ be generated only by the law-only joint lift and the prompt-HNL mass continuum. The source-malformed MEG-localized proposal and the old-nontransparent Warsaw/universal dipoles are excluded from the explicit survivor diagram. The pure finite-submatrix object is retained separately as a marginal U obstruction, not as a joint survivor.

The explicit equalizers are:

- the law-only object has its fixed base frame and joint H/U/L packet;
- two prompt-HNL objects are equal exactly when their threshold, oriented matching, complete H/U/L packet, and physical frame agree;
- law-only and prompt-HNL objects do not overlap because their frozen threshold/matching packets differ;
- every finite higher tuple is nonempty exactly when all retained base frames, threshold/matching records, H/U/L packets, and common-cone cocycles agree.

Theorem 16.1 (Codescent on the explicit survivor locus). *The codescent of $\mathcal{U}_{D3}^{\text{mark}}$ is the complete enriched standing-core groupoid of the executed law-only and prompt-HNL calibration locus. It contains continuum many connected base components. No same-base continuum multiplicity is asserted.*

Global classification boundary

The unbounded class of all B15 skeletons, all skeleton-pair equalizers, and all finite higher tuples is not solved here. The general overlap constructions are defined, but

$$\text{codesc}(\mathcal{U}_{D3}) \simeq \mathcal{N}(\hat{T}_{D3}^{\text{HUL}})$$

is not established by the present analysis. The corresponding global classification theorems remain open.

17 Failure and update structure

17.1 Total complement

For every raw lift, the ordered response system returns one status in $\{N/A, \text{Pass}, \text{Fail}, \text{Undef}, \text{ROOS}, \text{SU}, \text{TU}\}$. The complement of the survivor locus is the union of named source incoherence, physicality failure, response failure, undefined response, response-out-of-scope, acceptance failure, source update, target update, and inherited base complements.

17.2 No laundering

- A null direct search is not a no-HNL theorem.
- A nonunitarity bound is not exact unitarity.
- A CLFV null is not exact flavour conservation in every completion.
- Undefined or out-of-scope response is not zero and not pass.
- One failed presentation does not eliminate its base class.
- A new release, topology, detector primitive, or positive signal is target update.

Theorem 17.1 (Failure provenance on the evaluated loci). *Every evaluated marked empty locus or marked excluded parameter stratum carries an exact symbolic failure certificate identifying the first active failed relation, certified ROOS/N/A routing, or absence of a common H/U/L apex. A theorem covering every global empty fibre remains part of unresolved D3-11 and is not claimed here.*

18 Main theorem and physical interpretation

Theorem 18.1 (Main D3 continuum theorem). *At the frozen D3 target and source scope:*

1. *the prompt-HNL source family $\mathcal{R}_p$ consists entirely of one-point H/U/L survivors relative to the frozen ATLAS response specialization and has a positive mass/mixing applicability margin;*
2. *its retained base classes are pairwise SCIC-distinct, so $|\text{im } \rho_{D3}| \geq \mathfrak{c}$ and $|Q(\hat{T}_{D3}^{\text{HUL}})| \geq \mathfrak{c}$;*
3. *the pure finite-submatrix chart has an empty marginal U accepted locus on $\delta \in [0.02, 0.03]$, without complete-fibre elimination;*
4. *the Warsaw and universal electromagnetic dipoles are old-nontransparent and are not used for same-base splitting;*
5. *the protocol-indexed MEG-localized environment proposal is assigned SourceMalformed / NotReady before candidate formation and supplies no solved chart or codescent node;*
6. *exact same-base fibre multiplicity, global D3 fibre exhaustion, properness, global overlap closure, and complete PCR-5E handoff remain NotReady.*

18.1 Comparison with D1 and D2

D1 and D2 prove full-base preservation and universal fibre splitting at their declared scopes. D3 proves a continuum surviving base family and one exact marginal U obstruction. It does not establish same-base splitting and does not decide whether the complete D3 target preserves the full oscillation base or selects a proper subset.

18.2 Consequences and open questions

The prompt-HNL continuum proves that the D3 target does not select a unique oscillation standing-carrier class and that the enriched quotient remains continuum-sized. This differs from D1 and D2 in the coordinate tested, but not in producing a singleton: D1 constrains absolute scale, D2 constrains conjugation/LNV response, and D3 establishes a continuum of classes compatible with direct-state, active-closure, and CLFV records.

The present theorem does not determine whether every oscillation class admits a D3 lift, whether some complete base fibres are empty, or whether a fixed surviving base class splits into several enriched D3 classes. The pure finite-submatrix family provides a genuine U-chart obstruction, but chart failure alone is insufficient for complete-fibre elimination.

Principal conclusions

Statement	Conclusion	Status
Fixed target and source class	THUCL-D3-1.1 over PCR-3/B15	Established
Prompt-HNL joint survival	Continuum family on [28, 30] GeV	Established
Restriction image	$ \text{im } \rho_{D3} \geq \mathfrak{c}$	Established
Enriched quotient	$ Q(\hat{T}_{D3}^{\text{HUL}}) \geq \mathfrak{c}$	Established
Finite-submatrix U chart	Empty accepted locus on [0.02, 0.03]	Established marginally
Properness of the base image	Undetermined	Open
Fixed-base multiplicity	Undetermined	Open
Complete D3 atlas and joint handoff	Undetermined	Open

19 Falsifiability and direct counterexample criteria

The principal claims can be revised only by a counterexample that preserves the fixed target and comparison class while defeating a load-bearing implication. Representative possibilities are listed below.

Potential counterexample	Required construction	Affected result
Failure of transparency	A D3-only cell with a genuine old-target output on a chart classified transparent, or an inherited continuation that does not preserve the grading.	Retained transparency and restriction.
Incomplete source ontology	A same-target D3 source architecture outside the PCR-3 finite source sorts, operations, incidences, attachments, and finite hybrids.	Source coverage or the no-extension conclusion.

Potential counterexample	Required construction	Affected result
Spliced joint survivor	A purported H/U/L survivor whose records do not factor through one source-coherent physical apex.	One-point survival.
Prompt-response mismatch	A point in the prompt-HNL interval outside the declared ATLAS response specialization or outside an applicable U/L acceptance region.	Continuum survivor theorem.
Threshold-collapse counterexample	A lawful base-groupoid arrow identifying unequal physical thresholds while preserving the complete oriented matching packet.	Continuum base separation.
Finite-submatrix response error	An accepted pure finite-submatrix point on $\delta \in [0.02, 0.03]$, or failure of the public U chart's applicability on the complete pure source.	Marginal U obstruction.
Target change without retyping	A later search, CLFV channel, positive signal, or covariance revision used without a new target identifier.	Target identity and applicability.
Auxiliary classifier	A family name, naturalness criterion, matrix dimension, or preferred seesaw label that changes membership beyond the complete standing packet.	Same-scope standing identity.

20 Scope and revision conditions

20.1 Conditions that would revise the result

The lower-bound theorems would require revision if any of the following were established:

1. the target/source freeze omits an in-scope record or source architecture;
2. the publication-defined prompt width/lifetime map or ATLAS acceptance does not apply to a point of $\mathcal{R}_p$, or one operative U/L margin fails;
3. the pure finite-submatrix source does not satisfy its stated marginal U applicability assumptions or an accepted pure point exists on $[0.02, 0.03]$;
4. unequal explicit thresholds or complete H/U/L response packets are identified by an unlawful groupoid arrow;
5. a later empirical release or source primitive is admitted through an explicit target or source update.

20.2 Limitations

The theorem does not imply discovery of an HNL, a unique singlet count, a unique seesaw, exact unitarity, exact flavour conservation, CLFV independence, an exact global D3 atlas, complete-fibre finite-submatrix elimination, a proper base image, universal continuum splitting over every surviving base class, or a complete PCR-5E marginal

handoff. It does not identify a marginal U obstruction with base selection, a explicit chart construction with execution of the unbounded atlas, or a rejected source proposal with an admitted same-base survivor construction.

21 Implications for joint neutrino-carrier closure

The D3 calculation contributes three exact inputs to the later simultaneous neutrino-target analysis:

$$\mathcal{H}_{D3}^{\text{explicit}} = (\mathcal{R}_p, Q_{D3}^{\text{mark}}, \mathcal{U}_{D3}^{\text{mark}}, \text{Sep}_{\text{mark}}, \text{Fail}_{\text{mark}}).$$

They consist of the prompt-HNL continuum, the associated explicit restriction-image classes and overlap data, and the pure finite-submatrix marginal obstruction. These inputs establish that the D3 target admits continuum many base classes and that one specialized closure chart has an empty accepted interval.

They do not constitute a complete D3 marginal atlas. A full joint-target use requires the unresolved B15-indexed source classifications, pairwise and higher overlaps, complete support calculation, and fixed-base fibre analysis. In particular, the current D3 result neither proves a common generator with the symmetry target nor determines whether the complete joint neutrino target is singleton, finite, or continuously parameterized.

22 Conclusion

The heavy-state, active-closure, and charged-lepton-flavour extension does not collapse the oscillation standing-carrier space to a unique class. A source-first prompt-HNL family survives every applicable component of the fixed D3 target on one physical apex, preserves its retained oscillation core, and varies a standing-visible threshold and oriented matching datum across a real interval. The restriction image and enriched quotient are therefore both at least continuum-sized:

$$|\text{im } \rho_{D3}| \geq \mathfrak{c}, \quad \left| Q\left(T_{D3}^{HUL,1.1}\right) \right| \geq \mathfrak{c}.$$

The analysis also isolates a genuine active-closure obstruction on a pure finite-submatrix chart, while showing why that marginal failure cannot be promoted to elimination of an entire base class. Candidate same-base CLFV refinements either alter the old source-facing trace or fail source coherence at the present scope, so no fixed-base splitting theorem is inferred.

The result therefore locates D3's strongest established effect in direct-state and threshold/matching structure: the added target preserves continuum multiplicity while exposing nontrivial closure constraints. Determining the exact support of the D3 restriction image, the geometry of every lift fibre, and the relation of these coordinates to later cosmological and symmetry targets remains the next classification problem.

A Notation and formal objects

Symbol	Type	Meaning
H_a	B15 datum	Exact raw source domain, equations, applicability, status, survivor object, lawful groupoid, core map, packet moduli, provenance, deletion, failure, separators, and overlaps.
$Z_{a,X}$	solved object	Source/physical/response solutions in band X with all active predicates passed.
$\Gamma_{a,X}$	groupoid	Lawful source, frame, gauge/basis, chart, threshold, matching, and response presentation arrows preserving the complete packet.
I_{D3}	target-relative ideal	D3-only cells and finite closures with no old-target output.
R_{D3}	functor	In-situ old standing-core restriction after physical survival and routing.
$\mathcal{L}_X(q)$	groupoid	Homotopy fibre over the full connected base subgroupoid.
Q_{D3}^{sem}	semantic support	Base classes with a nonempty semantic one-point HUL fibre; exact support not classified here.
$Z_{U,\text{sub}}^{\text{pure}}$	pure-chart solution object	Empty accepted locus on the compact deficit interval; not a full base-fibre exclusion.

B Familywise response structure

Family	H chart	U chart	L chart	Joint/cross equations	Primary separators
LAW	Zero/direct process, nonlocal production, or ROOS by instrument.	Operational channel closure, normalization, leakage, memory, decay, and source/detector law.	Law/process dipole or exact zero.	Same law, channel, boundary, environment, and failure frame; no response-space splice.	Closure/leak, memory, operator order, CLFV process.
LOCAL	Resolved neutral states, masses, lifetimes, flavour attachments, production/decay.	Finite N, η, Θ only on applicable locus; generic boundary/closure otherwise.	HNL loop, dipole/contact, or zero.	Shared M, Y, Θ , lifetime, threshold, loop and matching data.	Mass/lifetime, rank, threshold, nonunitarity, CLFV.
HID	Hidden state, projection, visible decay, or no direct state.	Hidden support, dilation, projection, leakage, memory, active reduction.	Hidden-loop/effective transition or zero.	Same hidden ownership, projection, matching, vacuum, and common cone.	Hidden support, projection, leakage, memory.
COMP	Constituent production/decay response or zero.	Derived active closure with retained constituents.	Constituent/collective CLFV process.	Constituent retention and one source/physical apex.	Constituent support, threshold, response topology.
SOL	Profile excitation/decay response or zero.	Profile-mediated closure/continuation.	Localized profile CLFV transition.	Same profile, vacuum, boundary, and continuation.	Endability, profile support, closure, CLFV.
HYB	Arbitrary finite direct/hidden/boundary combinations.	Mixed closure, source/detector, hidden and EFT charts.	Interfering finite CLFV mechanisms.	Full finite higher incidence; pairwise agreement insufficient.	Joint/higher-arity, source composition, matching.

C Representative loci

Class/subatlas	D3 disposition	Proof witness	Diagnostic
q_W law-only Weinberg	Survives	No resolved H state, unitary operational U packet, and the source-defined light-neutrino CLFV response remains far below MEG II on one physical base point.	Null-search lift; no HNL selected.
$q_p(M)$, $M \in [28, 30]$ GeV	Continuum base family survives	Exact one-singlet source; exact membership at $c\tau_A = 0.2$ mm inside the frozen ATLAS lifetime-reweighting domain; positive lifetime and mass/mixing applicability margins; complete U chart table; exact L Pass; threshold/-matching separation.	Direct-state and threshold/matching base refinement.
Pure finite-submatrix chart	Marginal U rejection on $\delta \in [0.02, 0.03]$	Complete source point; applicable public submatrix chart; $N_e = 1 - \delta < 0.987$; H is ROOS.	Chart-level U obstruction only; no base elimination.
Environment-localized same-base route	Unresolved	No source-coherent environment law, complete common cone, MEG event kernel, or continuation audit is supplied.	No concrete same-base CLFV refinement theorem.

D Standing-coordinate summary

Coordinate	Exact witness	Relation to oscillation	PCR-5E tag
Finite-submatrix active deficit	Complete pure source, exact active block, trace-preserving hidden outcome, and applicable U row normalization.	Marginal U obstruction only; full-fibre selection unresolved.	closure-obstruction-submatrix
Prompt HNL mass/lifetime/topology	ATLAS prompt response specialization with $c\tau_A = 0.2$ mm, positive lifetime and mass/mixing applicability margins, and physical threshold/-matching packet.	Base-visible threshold/matching refinement with exact old restriction.	direct-state, rank-threshold
Operational U closure	Public oscillation closure plus the covariance-complete low-mass pion/kaon/tau LFU response; ORCA/global/EFT charts retain certified ROOS.	Retains source-specific closure rather than a universal matrix ontology.	active-closure-chart-complete
Environment-localized CLFV route	No admitted concrete witness; protocol-indexed support fails the source-coherence and physical-apex burden.	Same-base L refinement remains unresolved.	CLFV-same-base-open
Joint HNL H/U/L data	Shared mass, mixing, lifetime, threshold, loop, boundary, and flavour attachment.	Jointly unresolved outside the explicit survivor locus; may have P/Y source composition.	joint-synergy; no-common-generator-yet

E Dependency map

Source	Imported role	New D3 responsibility
PCR-2/PCR-4	Event-complete channel, shared witnesses, one-point survival, fail-closed statuses.	D3 H/U/L residual system and common physical apex.
PCR-3	Source-complete finite carrier ontology, activity reduction, finite hybrid closure.	D3 source-chart specialization; no new source extension.
PCR-5A/PCR-5B	Complete seven-channel SCIC packet, exact base atlas, B15 frames.	In-situ retained packet equality and stratumwise D3 lift atlas.
PCR-5C	Restriction, homotopy fibres, no merger, codescent, failure provenance.	Phenomenon-specific H/U/L realization and exact base image.
PCR-5D2	Physical amplitude realization, split-anchored one-point construction, in-situ transparency discipline.	D3 relative ideal, structural induction, and no deletion-defined restriction.
MGN5/MGN6	Matrix/rank/mixing standing and lawful quotient discipline.	D3 threshold, nonunitarity, boundary, and parameter descent.
Frozen external sources	Direct-search, nonunitarity, EW/flavour, EFT, and MEG II response predicates.	Exact applicability/acceptance solution objects; no unauthorized refit.
This paper	—	Pure finite-submatrix obstruction, source-first prompt-HNL continuum, complete U-chart disposition, and overlap geometry on the explicit survivor locus.

F Theorem synopsis

1. **D3-01/D3-02.** The target specification and semantic source coverage are established.
2. **D3-03/D3-04.** One-point physicality and in-situ retained transparency are proved on the explicit prompt-HNL chart; nontransparent points route by their actual packet.
3. **D3-05.** Semantic marginal and joint fibres are defined exactly; complete source-skeleton classification remains open.
4. **D3-06.** Parameter descent is established on the explicit charts.
5. **D3-07.** Exact no-repair source-composition profiles are obtained for the explicit base charts.
6. **D3-08.** $|\text{im } \rho_{D3}| \geq \epsilon$ and $|Q(\hat{T}_{D3}^{\text{HUL}})| \geq \epsilon$ from a genuine in-window H/U/L survivor continuum; no properness theorem is claimed.
7. **D3-09.** The proved diagnostic is a marginal finite-submatrix U obstruction and a prompt direct-state base continuum. The MEG-localized raw proposal is source-incomplete and excluded before candidate formation; no concrete same-base CLFV direction is closed.
8. **D3-10/D3-11.** The explicit base overlaps and evaluated failure frontiers are determined; global overlap/codescent and fixed-base splitting remain open.
9. **D3-12.** Only the explicit loci established here may be used in the subsequent joint-target analysis.

References

- [1] A. J. Maley, *Target-Fixed Physical Role Necessity for the Realized Neutrino Oscillation Interface*, PCR-1, 2026.
- [2] A. J. Maley, *Complete Physical Interface of the Neutrino Propagation Role Complex*, PCR-2, 2026.
- [3] A. J. Maley, *Exact Standing-Carrier Stratification of the Neutrino Oscillation Realization Space*, PCR-5B, 2026.
- [4] A. J. Maley, *Functorial Target Extension of Standing-Carrier Quotients*, PCR-5C, 2026.
- [5] A. J. Maley, *Lepton-Number-Violating Target Extension of the Neutrino Standing-Carrier Space*, PCR-5D2, 2026.
- [6] A. J. Maley, *PCR-5D3 Strengthened Detailed Theorem Skeleton*, 2026.
- [7] A. J. Maley, *PCR-5D3 Target Freeze and Source-Landing Closure*, 2026.
- [8] A. J. Maley, *Low-Mass Weak-Universality Target-Freeze Addendum*, TU-D3-1.1, 2026.
- [9] A. J. Maley, *Mass-Matrix, Eigenstructure, and Ratio Rigidity over the Canonical Coupling Carrier*, MGN5, 2026.
- [10] A. J. Maley, *Mixing-Angle and Phase Standing over Independently Standing Mass Carriers*, MGN6, 2026.
- [11] T2K Collaboration, *Search for heavy neutrinos with the T2K near detector ND280*, 2019.
- [12] NA62 Collaboration, *Search for heavy neutral lepton production in K^+ decays to positrons*, 2020.
- [13] NA62 Collaboration, *Search for K^+ decays to a muon and invisible particles*, 2021.
- [14] Belle Collaboration, *Search for a heavy neutral lepton that mixes predominantly with the tau neutrino*, 2024.
- [15] CMS Collaboration, *Search for long-lived heavy neutral leptons with lepton flavour conserving or violating decays to a jet and a charged lepton*, 2024.
- [16] CMS Collaboration, *Search for long-lived heavy neutral leptons in proton-proton collision events with a lepton-jet pair associated with a secondary vertex*, 2025.
- [17] ATLAS Collaboration, *Search for heavy neutral leptons in decays of W bosons using displaced vertices*, 2025.
- [18] NA62 Collaboration, *Search for heavy neutral leptons in π^+ decays to positrons*, 2025.
- [19] LHCb Collaboration, *Search for heavy neutral leptons in beauty-hadron decays*, 2025.
- [20] ATLAS Collaboration, *Search for heavy neutral leptons in prompt W -boson decays*, 2025.
- [21] M. Blennow et al., *Bounds on lepton non-unitarity and heavy neutrino mixing*, 2023.
- [22] M. Blennow et al., *Misconceptions in Neutrino Oscillations in presence of a non-Unitary Mixing*, 2025.
- [23] O. Kozynets et al., *Constraints on non-unitary neutrino mixing in light of atmospheric and reactor neutrino data*, 2025.
- [24] KM3NeT Collaboration, *Study of tau neutrinos and non-unitary neutrino mixing with ORCA6*, 2025.
- [25] A. Broncano, M. B. Gavela, and E. Jenkins, *The effective Lagrangian for the seesaw model of neutrino mass and leptogenesis*, 2003.
- [26] A. Broncano, M. B. Gavela, and E. Jenkins, *Neutrino physics in the seesaw model*, 2003.
- [27] MEG II Collaboration, *New limit on the μ^+ to e^+ gamma decay with the MEG II experiment*, 2025.
- [28] A. Ilakovac and A. Pilaftsis, *Flavour-violating charged lepton decays in seesaw-type models*, 1995.
- [29] B. Grzadkowski, M. Iskrzynski, M. Misiak, and J. Rosiek, *Dimension-Six Terms in the Standard Model Lagrangian*, JHEP 10 (2010) 085.

- [30] D. Bryman, V. Cirigliano, A. Crivellin, and G. Inguglia, *Testing Lepton Flavor Universality with Pion, Kaon, Tau, and Beta Decays*, 2022.
- [31] E. E. Jenkins, A. V. Manohar, and P. Stoffer, *Low-Energy Effective Field Theory below the Electroweak Scale: Operators and Matching*, 2018.